AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM351
Data management and analysis

Final Examination
Spring Semester 2024/2025

Date: 28/June/2025

Number of Exam Pages: (4)	Time Allowed: (3) Hours

1- This exam has 2 parts.
2- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
3- Student handbooks are NOT permitted in this examination.
4- The use of electronic devices that could have a memory is NOT permitted.
5- At the end of the examination, check that you have written your student ID, name and your section number on the first page.

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Part One: [10 marks]

Please select **the best and most appropriate answer** and write it on the answer booklet.

1) What is the primary purpose of a database management system (DBMS)?
   a) To provide a platform for creating computer games
   b) To facilitate the creation and maintenance of a database
   c) To provide an interface for users to interact with a database
   d) To provide a platform for video editing software

2)Which of the following is NOT a type of database model?
   a) Hierarchical
   b) Relational
   c) Object-Oriented
   d) Linear

3) What is the purpose of normalization in database design?
   a) To increase data redundancy
   b) To decrease data consistency
   c) To increase data storage space
   d) To decrease data redundancy

4) Which of the following is NOT a type of SQL join?
   a) Inner Join
   b) Outer Join
   c) Left Join
   d) Center Join

5) Which of the following is a disadvantage of NoSQL databases?
   a) They have strict schema requirements
   b) They are not scalable
   c) They lack support for ACID transactions
   d) They are not suitable for large dataset

6) What is the purpose of data modeling in database design?
   a) To create a graphical representation of a database
   b) To identify the relationships between data entities
   c) To ensure data consistency and accuracy
   d) To establish data security protocols

7) Which of the following is NOT a property of a transaction in database management?
   a) Atomicity
   b) Consistency
   c) Durability
   d) Velocity

8) Which of the following is a benefit of using a distributed database system?
   a) Increased data consistency
   b) Reduced network traffic
   c) Decreased data availability
   d) Increased data fragmentation

9) Which of the following is NOT a type of database backup?
   a) Full backup
   b) Differential backup
   c) Incremental backup
   d) Virtual backup

10) Which of the following is a primary function of a database administrator (DBA)?
   a) Developing software applications
   b) Ensuring data security and privacy
   c) Creating network topologies
   d) Providing customer support services

Part Two: [90 marks]

Question One: [30 marks]

The library of Arab Open University Muscat branch stores information about books, members, borrows, categories, reservations, fines, and reviews in AOU_LIBRARY database.

The database schema of AOU_LIBRARY is given below: -

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1. Book table: contains information about the books available in the library, such as **bookID**, title, authorName, publisherName, publicationDate, *categoryID* and ISBN. **bookID** is the primary key of the table. *categoryID* could be included as foreign key.
2. Member table: contains information about the members of the library, such as **memberID**, firstName, lastName, address, phone, email, membershipFees, and membershipType. **memberID** is the primary key of the table.
3. Borrow table: contains information about the book borrows made by members, such as **borrowID**, *bookID*, *memberID*, borrowDate, and returnDate. **borrowID** is the primary key of the table. *bookID* and *memberID* columns could be included as foreign keys.
4. Category table: contains information about the categories of books available in the library, such as **categoryID** and categoryName. **categoryID** is the primary key of the table.
5. Reservation table: contains information about the reservations made by members for books, such as **reservationID**, *bookID*, *memberID*, and reservationDate and reservationStatus. **reservationID** is the primary key of the table. *bookID* and *memberID* columns could be included as foreign keys.
6. Fine table: contains information about the fines charged to members for late returns, such as **fineID**, *memberID*, *bookID*, fineAmount, fineDate and paymentStatus. **fineID** is the primary key of the table. *bookID* and *memberID* columns could be included as foreign keys.
7. Review table: contains information about the reviews left by members for books, such as **reviewID**, *bookID*, *memberID*, rating, and reviewText. **reviewID** is the primary key of the table. *bookID* and *memberID* columns could be included as foreign keys.

Use the above relations and write SQL statements for the following questions: -

A	Retrieve the title, author name and reservation date of all books reserved by a member with member ID "AOU_456" and reservation status "CONFIRMED".	[5 marks]
B	Retrieve the title and number of books that have been borrowed more than 5 times.	[6 marks]
C	Retrieve the count of all members who have borrowed books, grouped by membership type.	[4 marks]
D	Increase the fine amount for all members of membershipType 'student'. (*Hint: use subquery*)	[5 marks]
E	Retrieve the average rating and number of reviews for each book in the Reviews table. Include only books that have at least 3 reviews and an average rating of 4.0 or higher	[6 marks]
f	Retrieve the title and author of each book, along with the category name. Include only the books in the category "PHP Programming".	[4 marks]

Question Two: [20 marks]

Use the sample below from the hall document database collection to answer the following questions: (assume the database name is: "db" and the collection name is: "hall").

```
{"hall_code": "H01",
"details": {
"hall_name": "Party Center",
"rent": 150,
"capacity": 500
```


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```
},
"booking": [
{"booking_id": 1, "customer_id": 111, "email":marwan@gmail.com, booking_for_date:
"ISODate("2022-12-09T08:00:00z")" },
{"booking_id": 5, "customer_id": 777, "email":khalid@gmail.com, booking_for_date:
  "ISODate("2022-12-20T08:00:00z")" }
]
}
```

a. Find all documents in the hall collection. [2 marks]
b. Find any document from the hall collection where the rent is 100. [4 Marks]
c. Find the booking details of customer with id 777. [4 marks]
d. Find the number of halls having capacity of 500 in the collection. [5 marks]
e. Find all the details of halls where the booking for date is before 30-12-2022. [5 marks]

Question Three: [40 marks]

a. Convert the following JSON representation to XML [10 marks]

```
{ "Image":
 { "Width": 800,
 "Height": 600,
"Title": "View from 15th Floor",
"Thumbnail": {
 "Url": "http://www.example.com/image/481989943", "Height": 125, "Width": 100
},
"Animated" : false,
"IDs": [116, 943, 234, 38793]
}
}
```

b. Which are the two different types of Views in SQL? Explain. [5 marks]
c. Which are the two main aspects of Data Integration? Explain. [5 marks]
d. Distinguish between data querying and data mining: [5 marks]
e. Can normalization avoid 100% redundancy? Please justify your answer. Which are the components of Data Warehousing Systems? [5 marks]
f. An experiment involves two groups of participants, Group A and Group B and we have recorded the grades of each group member for two courses, M251 and TM351 as follows:

Group A

StudentID	M251	TM351
A1	75	80

Group B

StudentID	M251	TM351
B1	60	70

Assume that a new student, S1 whose grade of M251 is 62 and grade of TM351 is 75 who needs to be placed into either Group A or Group B. Calculate the similarity ratio of S1 with A1 and B1. (10 marks)

End of Questions

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